

# ME6102 Presentation 3

# Automatic Chess Board

—

Khanjan Parikh  
Arkapravo Patra  
Udayan Gupta  
Tarshit Sehgal

# Fabrication Details + Contribution

- **Procured metal parts:** Extrusion rails (aluminium) , 8mm steel rods, linear and radial bearings - **Udayan + Khanjan**
- **3D Printing:**
  - Most motor, bearing connectors, ie, gantry, was 3d printed from PLA. Infill was made **50%**, wall loops - **3/4** (according to function) with **tri hexagonal** pattern for best strength capabilities. - **Arkapravo + Khanjan**
  - Stress bearing parts were reinforced with additional PLA growths, to improve lateral stability and increase overall moment of inertia. - **Tarshit**
  - Play of the steel rods and linear bearings were reduced with friction fit bolted parts - **Arkapravo**
- **Laser Cutting**
  - Laser Cutting and engraving were done for the main chess board. - **Tarshit + Khanjan**
  - Planar parts like bearing placeholders (functions as retaining rings) were laser cut from scrap acrylic to reduce costs - **Udayan**

# Rudimentary Analysis

$$F \propto \frac{1}{d^2}$$

$$\frac{F_2}{F_1} = \left( \frac{d_0}{d_0 + t} \right)^2$$

$d_0 \rightarrow$  dist b/w object  
and magnet without sheet

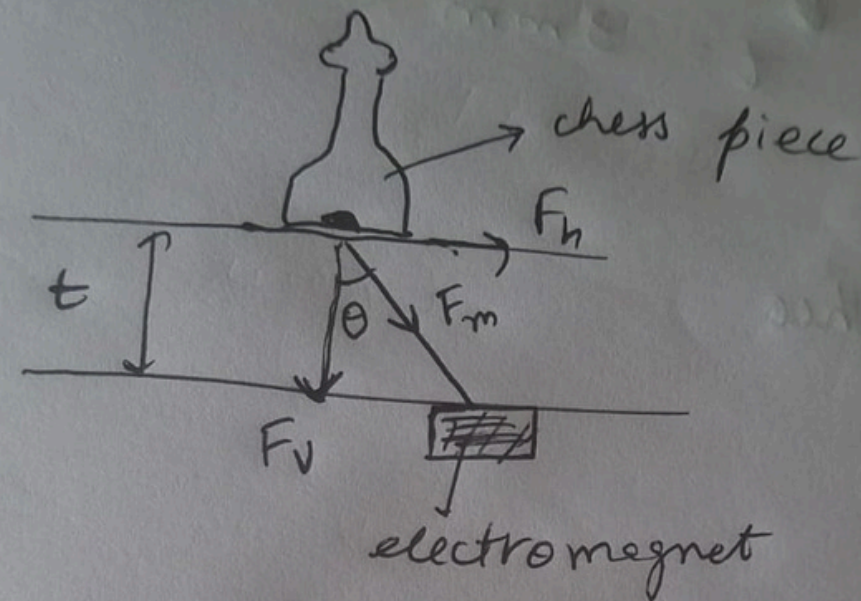
let  $F_m \rightarrow$  magnetic force,  $F_0 \rightarrow$  rated force  
of electromagnet

$\eta = \frac{1}{2}$  (safety factor),  $t \rightarrow$  thickness of  
acrylic sheet (3mm)

$$F_m = F_0 \left( \frac{d_0}{d_0 + t} \right)^2$$



# Rudimentary Analysis



$$F_h \geq \text{frictional force}$$

$$F_h \geq \mu(mg + F_v)$$

$$F_m (\sin \theta - \mu \cos \theta) \geq \mu mg$$

$$F_0 \left( \frac{d_0}{d_0 + \frac{t}{\cos \theta}} \right)^2 (\sin \theta - \mu \cos \theta) \geq \mu mg$$



# Rudimentary Analysis

$$F_0 \geq \frac{\mu m g}{(\sin \theta - \mu \cos \theta) \left( \frac{d_0}{d_0 + \frac{t}{\cos \theta}} \right)^2}$$

$$F_{0 \min} = \eta \times \frac{\mu m g}{(\sin \theta - \mu \cos \theta) \left( \frac{d_0}{d_0 + \frac{t}{\cos \theta}} \right)^2}$$

safety factor (=2)

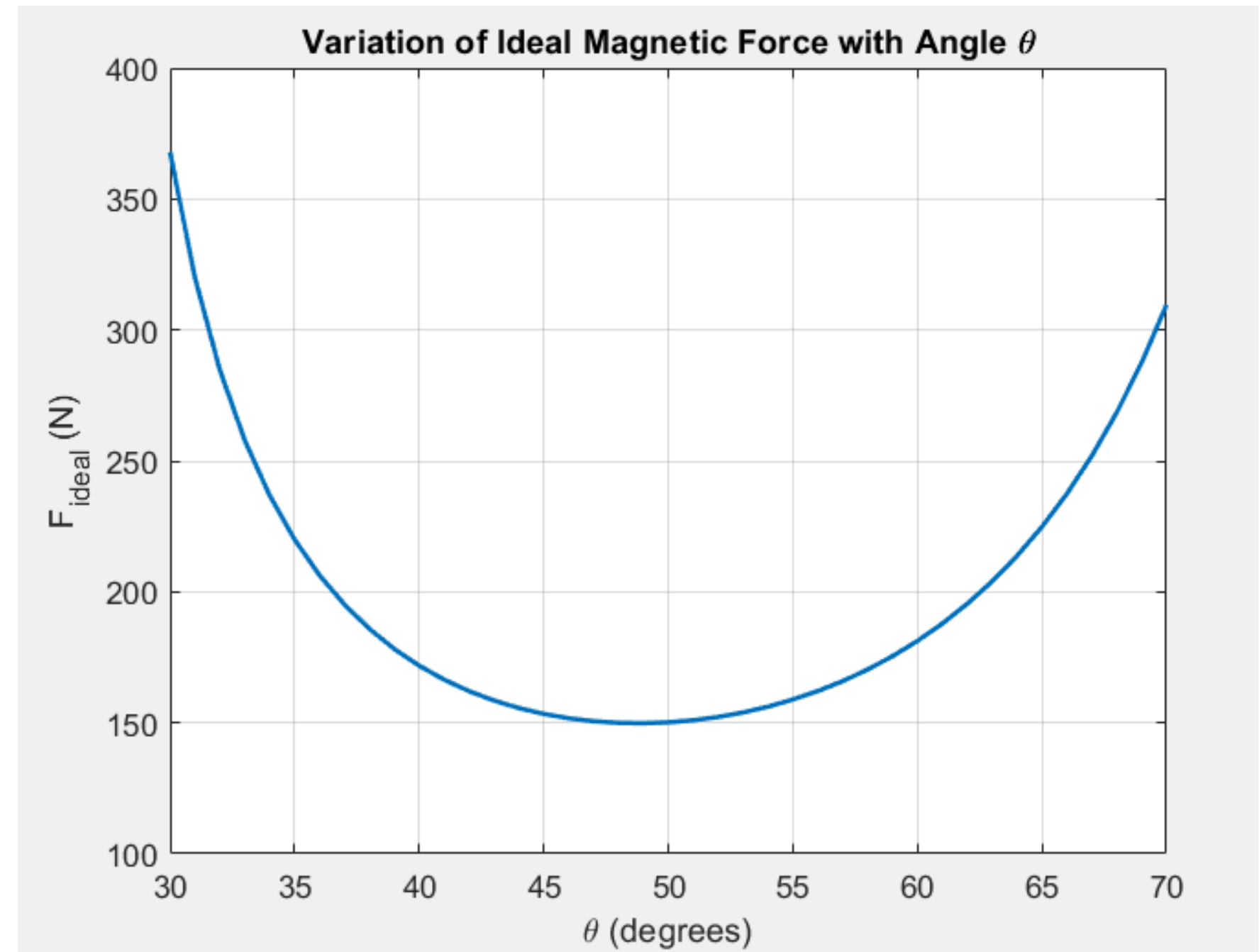
using  $\rightarrow$

$$\begin{aligned}\mu &= 0.45 \\ m &= 0.025 \text{ kg} \\ g &= 9.8 \text{ m/s}^2 \\ d_0 &= 0.2 \text{ mm} \\ t &= 3 \text{ mm}\end{aligned}$$

# Rudimentary Analysis

## But what about theta?

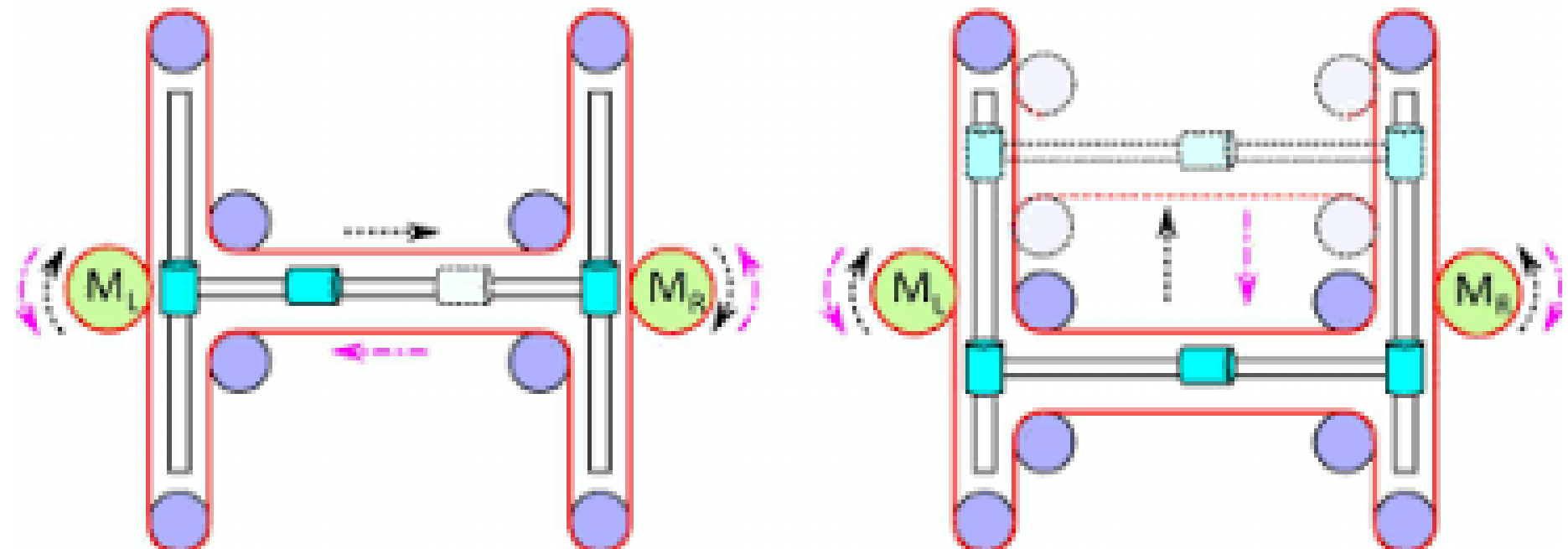
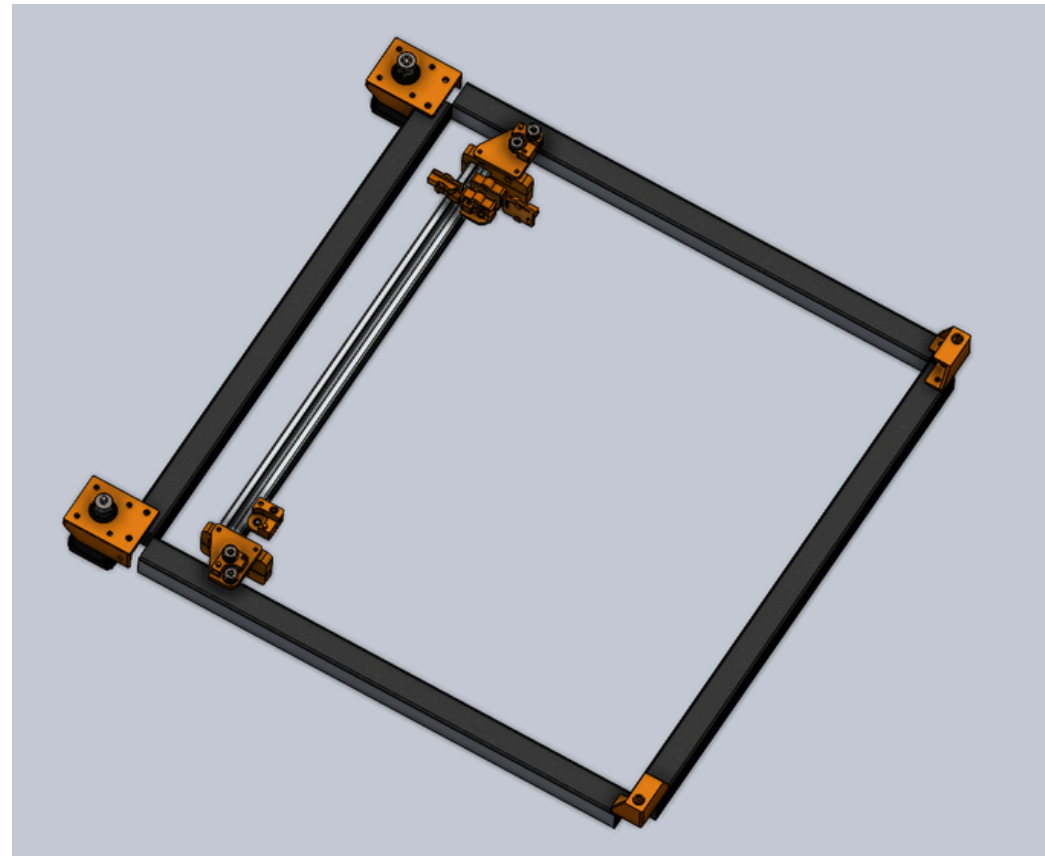
Theta was varied from 30 degrees to 70 degrees and a minima occurred at 49 degrees which gave actuator force to be 150N, with safety factor of two, it became **300N**



| Component       | Specification                        | Status   | Mathematical Justification for Selection                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------|--------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NEMA 17 Stepper | 1.8°/step, 4.4 kg-cm holding torque  | Received | Torque Requirement Analysis:<br>Estimated Gantry Mass (M): 0.8 kg<br>Desired Max Acceleration (a): 0.5 m/s²<br>Frictional Force (F_f): ~6.0 N (estimate)<br>Required Force (F_req) = M*a + F_f = (0.8 * 0.5) + 6.0 = 6.4 N<br>GT2 Pulley Radius (r): 8 mm = 0.008 m<br>Required Torque (τ) = F_req * r = 6.4 N * 0.008 m = 0.0512 Nm = 0.56 kg-cm<br>The selected motor's 4.4 kg-cm torque provides a safety factor of ~7.8, which is robust for overcoming inertia and friction.   |
| Electromagnet   | 12V DC, 35kg Lifting Force           | Ordered  | Done in previous slides!                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| A4988 Driver    | 1/16 Microstepping, Current Limiting | Received | Control Resolution & Motor Protection:<br>The L298N driver was initially considered but rejected. It lacks current limiting, risking motor burnout, and is limited to full steps (1.8°).<br>The A4988 allows for 1/16 microstepping, increasing resolution from 200 to 3200 steps/revolution. This is critical for smooth, quiet, and precise motion. Its built-in potentiometer allows setting the coil current to the motor's 1.2A rating, maximizing torque without overheating. |
| GT2 Timing Belt | 2mm pitch                            | Received | Motion Transmission: Standard for high-precision applications, providing zero-backlash power transmission essential for repeatable positioning.                                                                                                                                                                                                                                                                                                                                     |
| TIVA TM4C123G   | ARM Cortex-M4 @ 80MHz, FPU           | Received | Processing Power: The ARM Cortex-M4 core with a hardware Floating-Point Unit (FPU) is ideal for efficiently handling the kinematic transformations and Bresenham's line algorithm calculations in real-time without performance loss. Ample timers ensure precise step pulse generation.                                                                                                                                                                                            |

# Mechanical Design: From CAD to Reality

- **Design Philosophy:** The system is designed as an H-Drive gantry. This mechanism was chosen for its mechanical simplicity, using only two stationary motors and a single continuous belt loop, minimizing the moving mass and thus the required motor torque compared to Cartesian designs where motors move on the axes.





# System Force and Torque Calculation

This calculation determines the minimum required motor torque based on the **most demanding** operational conditions: including moving a heavy 200g chess piece (m) with a rapid acceleration of 1.0 m/s<sup>2</sup> (a).

| Parameter                         | Scenario A1          | Scenario A2          | Scenario B1          | Scenario B2          |
|-----------------------------------|----------------------|----------------------|----------------------|----------------------|
| <b>Input Parameters</b>           |                      |                      |                      |                      |
| Chess Piece Mass                  | 100 g                | 100 g                | 200 g                | 200 g                |
| Target Acceleration               | 0.5 m/s <sup>2</sup> | 1.0 m/s <sup>2</sup> | 0.5 m/s <sup>2</sup> | 1.0 m/s <sup>2</sup> |
| <b>Force Calculations (N)</b>     |                      |                      |                      |                      |
| Frictional Force ( )              | 0.294 N              | 0.294 N              | 0.589 N              | 0.589 N              |
| Inertial Force ( )                | 0.789 N              | 1.577 N              | 0.839 N              | 1.677 N              |
| <b>Total Linear Force ( )</b>     | <b>1.083 N</b>       | <b>1.871 N</b>       | <b>1.428 N</b>       | <b>2.266 N</b>       |
| <b>Torque Calculations (N·cm)</b> |                      |                      |                      |                      |
| Dynamic Torque ( )                | 6.90 N·cm            | 11.91 N·cm           | 9.09 N·cm            | 14.42 N·cm           |
| Safety Factor ( )                 | 2.0                  | 2.0                  | 2.0                  | 2.0                  |
| <b>Final Specified Torque ( )</b> | <b>13.8 N·cm</b>     | <b>23.8 N·cm</b>     | <b>18.2 N·cm</b>     | <b>28.8 N·cm</b>     |

# Electromagnet Requirements

This calculation determines the minimum required pulling force based on the **most demanding** operational conditions: including moving a heavy 200g chess piece (m) with a rapid acceleration of 1.0 m/s<sup>2</sup> .

**Assumptions :**

1 kgf = 9.81 N

Safety Factor S=3

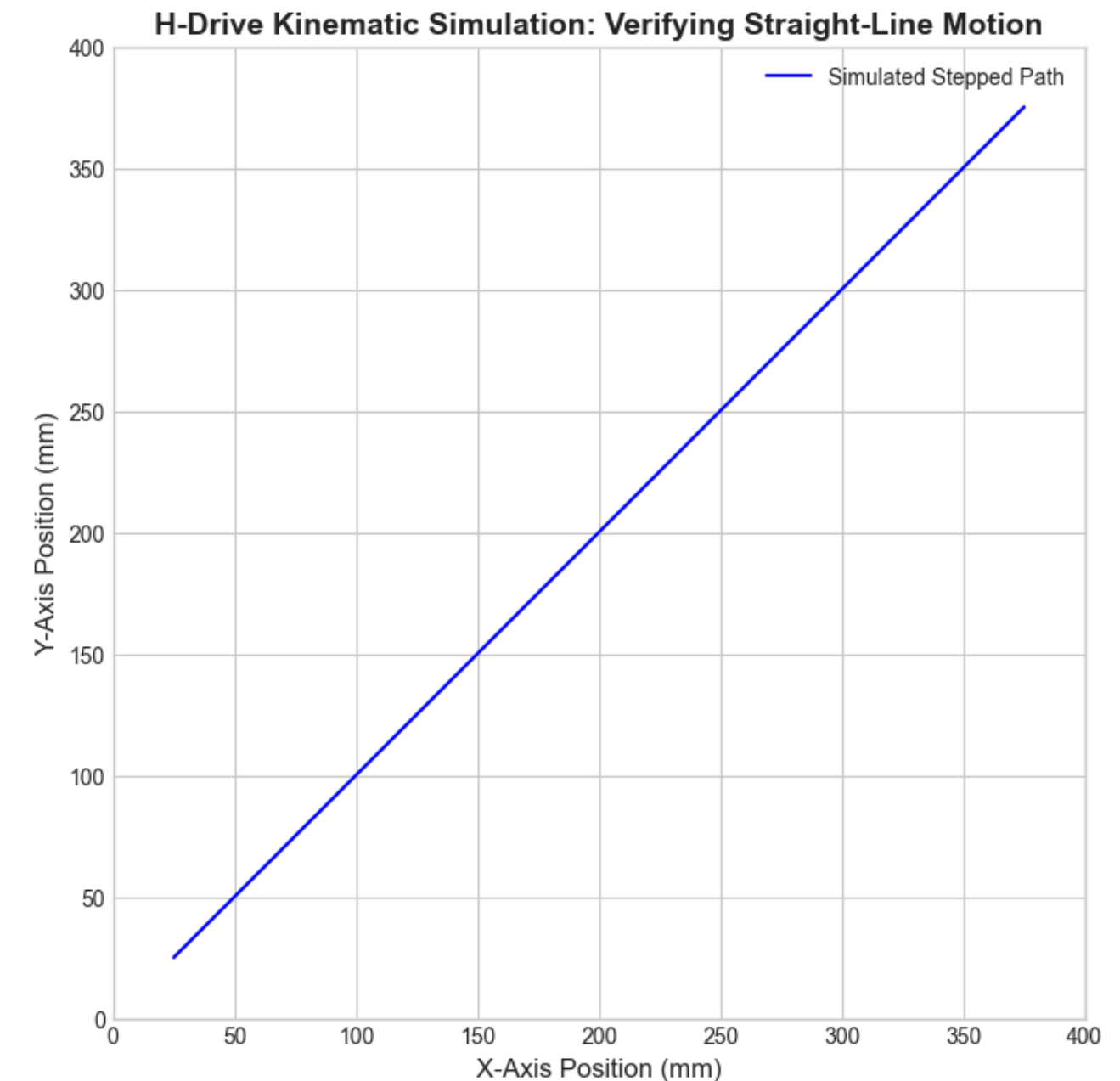
Gap efficiency η=0.05

| Scenario             | Description                | Working Force (N) | Design Force (N) [S=3] | Required Contact Pull (N) | Required Contact (kgf) |
|----------------------|----------------------------|-------------------|------------------------|---------------------------|------------------------|
| A — Hold piece       | Just hold 28 g chess piece | 0.275             | 0.824                  | 16.48                     | 1.68                   |
| B — Low move         | Low dynamic load (2 N)     | 2                 | 6                      | 120                       | 12.23                  |
| C — Recommended move | Conservative motion (5 N)  | 5                 | 15                     | 300                       | 30.58                  |
| D — Heavy / extreme  | Extreme case (10 N)        | 10                | 30                     | 600                       | 61.16                  |

From calculations, a magnet rated ≥30 kgf contact pull satisfies our 15 N design requirement through acrylic + air gap (5% efficiency). Hence, 35 kgf magnet (12 V, 0.34 A) is selected — adequate for our motion scenario.

# H-Drive Kinematic Simulation

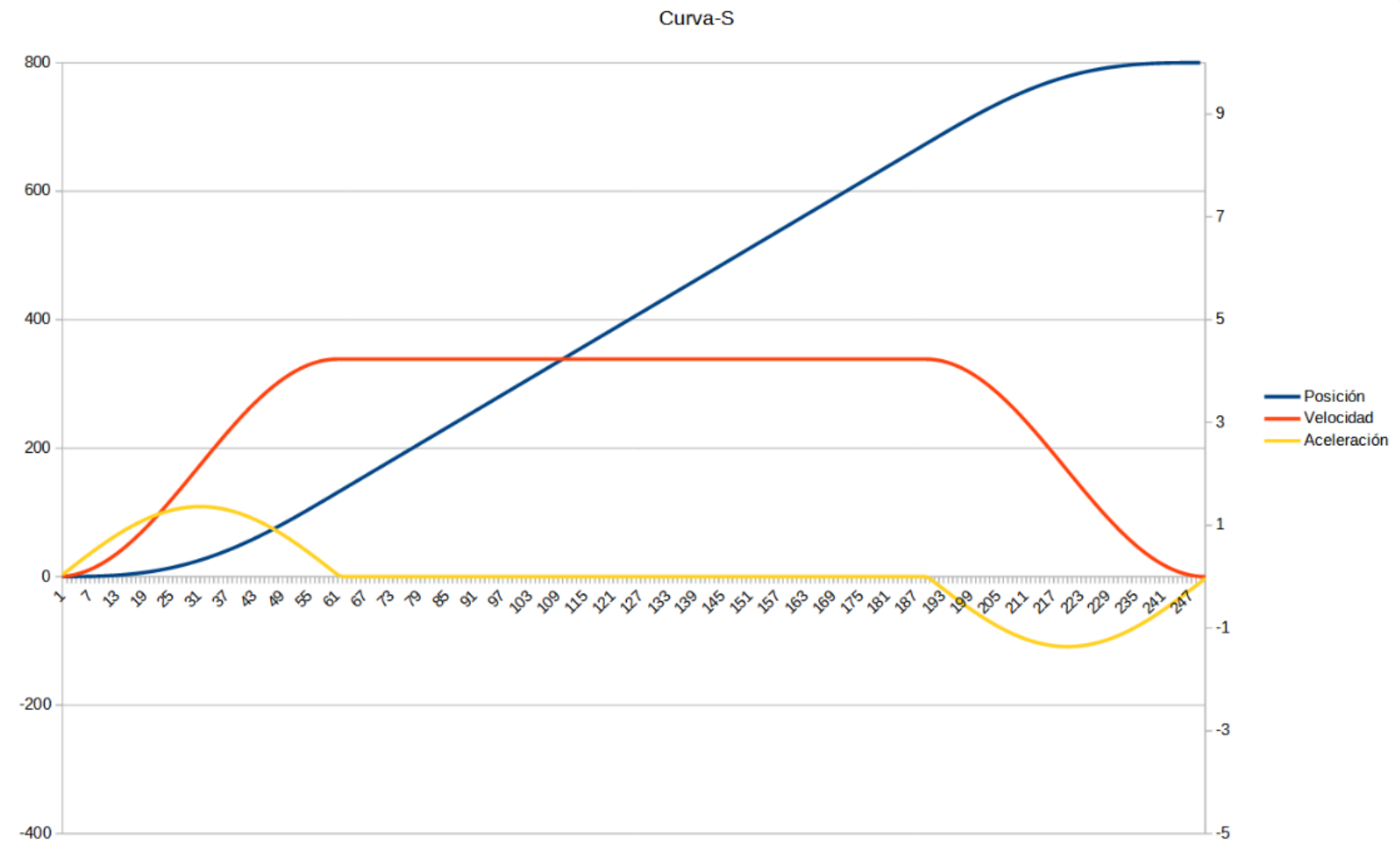
- To verify our H-Drive kinematic equations and control logic, we developed a Python simulation. This script models the exact process the TIVA microcontroller will follow.
- First, we convert the start and end chess squares—in this case, 'a1' to 'h8'—into precise physical coordinates in mm, and then into discrete machine steps.
- Next, we use **Bresenham's Line Algorithm**. This is the same algorithm implemented on the microcontroller, which intelligently decides which motor to pulse at each moment to create the straightest possible line.
- Finally, we plot the results. The blue solid line shows the path constructed from thousands of tiny, discrete motor steps generated by our algorithm.
- This simulation gives us high confidence that our embedded code will produce the smooth, precise, and straight-line motion required for the chessboard before we test it on the final hardware.





# Motion planning

- Possible issue: electromagnet running too fast with respect to chess piece
- Maximum acceleration possible using 343N actuator was found out to be  $6.6\text{m/s}^2$
- Using S-curve to avoid jerks
- Maximum possible acceleration is  $6.6\text{m/s}^2$ , will have to limit the acceleration below this number



# Big plan

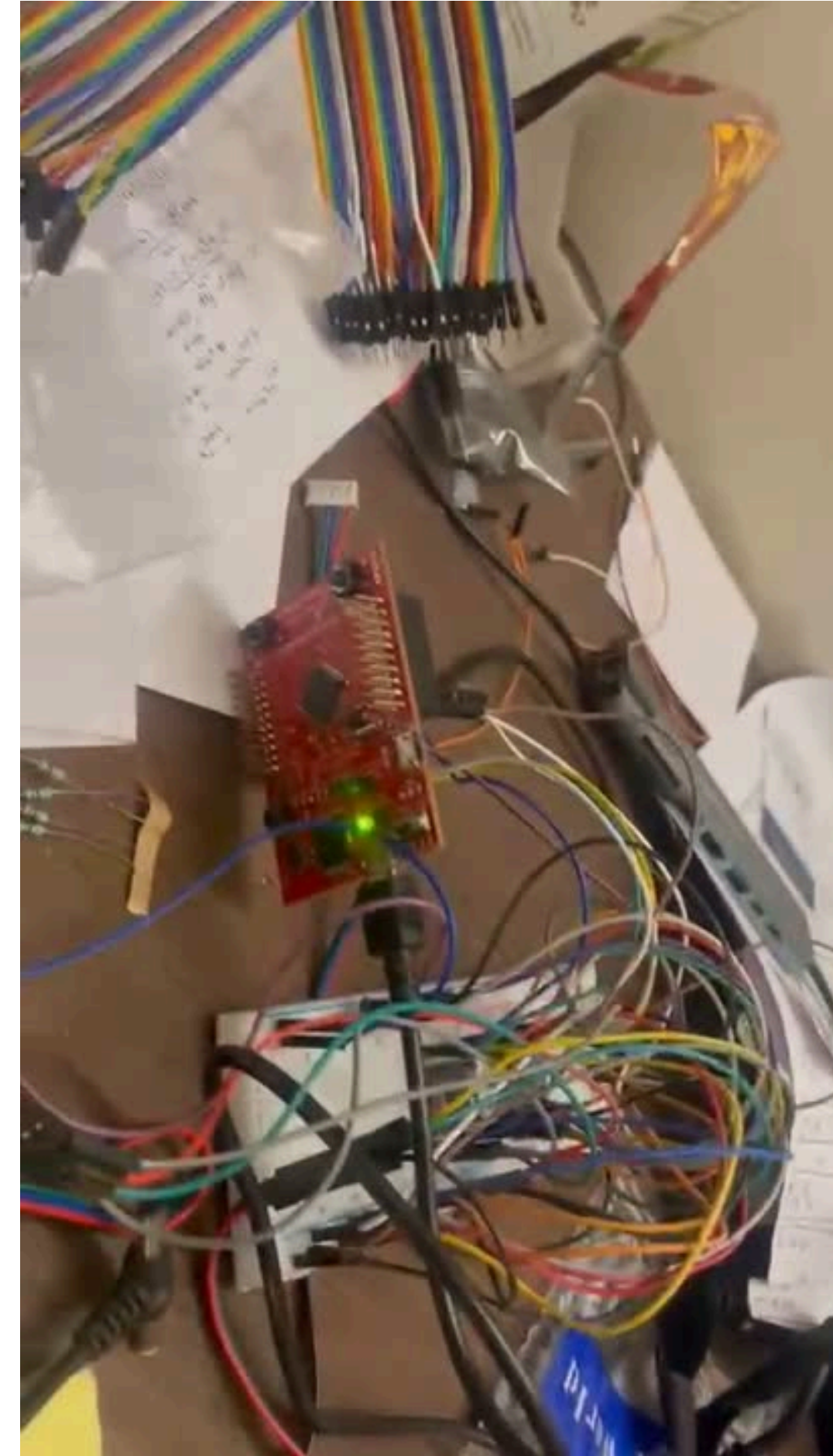
- The plan was to make an automated chess board, which can play against a human, all by itself
- After mechanical assembly and successful movement of a single chess piece kept alongside others, without magnetic interference or interaction of surrounding chess pieces, the idea was to integrate hall sensors to detect the current position of all the pieces on chess board.
- This is possible if we have 64 hall sensors, one for each box. Since we know initial positions of each piece, using this information, the current state of the board can be found.

# How to use?

## How to Set Position Coordinates

- In the provided code `tiva_corexy_magnet.c`, you control movement by calling the `moveTo(x, y)` function inside `main()`.
- Coordinate System: The board uses a millimeter (mm) grid.
- (0, 0) is your starting "Home" corner.
- (X, Y) is the target position in millimeters.
- How to Edit Moves: Look for the `// --- Control Sequence ---` section inside `main()`. You will see lines like this:
- To change where the piece goes: Simply change the numbers in `moveTo(...)`.
- Example: To move to the center of a 400mm board: `moveTo(200.0f, 200.0f);`
- Example: To move 10mm right: `moveTo(10.0f, 0.0f);`
- Important: The code keeps track of its current position. If you are at (50, 20) and call `moveTo(60, 20)`, it will move +10mm in X. It is an absolute positioning system, not relative.
- 3. The Complete Sequence
- Your `main()` loop currently runs this sequence automatically:
- Wait 1 second (Safety pause).
- Turn Magnet ON (Grabs the piece).
- Move to (50, 20) (Slides piece to this coordinate).
- Turn Magnet OFF (Drops the piece).
- Move back to (0, 0) (Returns the empty magnet to start).

# Working Videos - Sensor working





# Final Demo



# Initial Testing



# Unmet Goals

- Could not make it fully functional ready to use, instead, we just provided proof of concept. Reasons: Due to budgetary and time constraints. We wanted to use 64 hall sensors and each hall sensor costs 30INR, but since we had to restrict ourselves under 5000 INR, we bought just few to test our concept.
- Most of the parts used in our project were 3D-printed, which turned out to faulty many a times when we used micro-factory facilities, this wasted a lot of our time in just removing supports or filing the parts.
- And last, getting everything to work out was difficult than we thought, as we faced many small issues throughout the journey. So, the time we could give to project was less than what was required for its completion.

# Learnings from the project

- Due to overheating of the stepper motor driver, it used to go to wrong positions, even if correct command was given. We observed that it happened only after the system was on and working for sometime. This was big learning, as we never thought overheating can drastically effect performance like this
- The biggest lesson learned was on tolerances, many a times we had to file a part to reduce the external or increase the hole diameter
- Digitization of system to write the code was another big thing which we learned

